

KUEMMERSBRUCK, Germany

WMO: 107710

Lat: 49.4283N

Lon: 11.9017E

Elev: 1371

StdP: 13.98

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	13.8	4.9	7.5	9.3	10.0	9.6	15.7	23.9	41.6	20.7	41.2	2.6	100	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	87.4	66.2	83.7	64.9	80.2	63.8	67.9	82.4	66.4	80.1	64.9	77.2	5.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	91.0	70.8	61.7	86.5	70.1	60.3	82.3	69.5	33.0	82.3	31.7	80.2	30.6	77.6	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
18.0	15.1	12.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			1.6	92.2	6.4	3.2	-3.0	94.5	-6.7	96.4	-10.3	98.2	-14.9	100.5
			2.1	70.6	6.5	2.3	-2.6	72.2	-6.3	73.5	-10.0	74.8	-14.6	76.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	29.8	32.7	39.4	47.8	56.2	62.2	65.0	64.5	56.5	47.7	38.4	31.7	(6)
	DBStd	14.41	8.31	8.14	7.35	7.20	6.69	6.91	6.18	5.80	6.01	6.88	6.33	7.70	(7)
	HDD50	2643	627	485	332	123	18	1	0	0	12	128	350	567	(8)
	HDD65	6509	1092	905	793	516	283	135	79	79	260	537	798	1032	(9)
	CDD50	1821	0	0	4	58	211	367	465	451	207	56	2	0	(10)
	CDD65	211	0	0	0	0	10	51	79	65	6	0	0	0	(11)
	CDH74	2602	0	0	0	16	191	599	886	803	106	1	0	0	(12)
	CDH80	803	0	0	0	1	38	180	309	262	13	0	0	0	(13)
(14) Wind	WSAvg	5.0	5.3	5.9	5.9	5.2	5.0	4.8	5.0	4.6	4.4	4.3	4.6	5.1	(14)
(15) Precipitation	PrecAvg	29.6	2.4	2.0	2.0	1.7	2.7	3.1	3.2	2.8	2.4	2.2	2.5	2.6	(15)
	PrecMax	43.7	5.3	5.4	5.6	3.6	6.3	5.8	5.2	5.7	6.4	7.8	5.7	7.2	(16)
	PrecMin	20.3	0.3	0.3	0.4	0.4	0.7	0.8	0.7	0.9	0.9	0.6	0.0	0.5	(17)
	PrecStd	4.6	1.3	1.3	1.3	0.8	1.3	1.3	1.3	1.2	1.2	1.4	1.2	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.4	54.5	64.5	75.8	84.5	89.8	91.7	94.1	81.8	71.0	57.6	50.6	(19)
		MCWB	46.3	46.1	50.3	56.3	63.3	67.1	67.5	66.9	63.1	59.9	51.5	46.9	(20)
	2%	DB	47.9	49.7	60.0	71.5	79.2	85.1	87.6	86.2	77.2	65.2	53.0	47.6	(21)
		MCWB	44.2	43.6	48.6	55.6	61.8	65.9	66.3	65.2	62.9	56.7	48.8	44.4	(22)
	5%	DB	44.8	46.8	55.3	67.0	74.6	81.3	83.3	82.1	73.1	61.5	50.0	44.1	(23)
		MCWB	41.5	42.3	46.8	53.1	59.2	64.3	65.1	64.1	60.3	55.0	46.5	41.5	(24)
	10%	DB	41.1	44.0	51.0	62.6	70.3	76.7	78.7	78.3	69.0	58.1	47.2	41.2	(25)
		MCWB	38.6	40.1	45.0	51.0	57.0	62.1	63.8	63.2	58.8	52.9	44.5	39.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	47.1	52.2	58.9	65.5	69.3	69.6	69.9	66.7	60.6	52.2	47.5	(27)
		MCDB	50.4	52.6	60.5	73.0	81.1	85.2	85.6	85.8	76.9	68.1	55.9	49.9	(28)
	2%	WB	44.4	44.6	49.9	56.1	62.6	67.2	67.9	67.2	63.6	57.9	49.4	44.7	(29)
		MCDB	47.4	48.9	57.6	69.5	77.1	81.4	82.7	81.2	73.5	63.9	52.5	47.3	(30)
	5%	WB	41.7	42.5	47.7	53.8	60.3	65.3	66.3	65.6	61.8	55.6	47.0	41.7	(31)
		MCDB	44.6	46.4	54.1	65.3	72.3	78.3	80.1	78.6	70.7	60.5	49.5	43.9	(32)
	10%	WB	38.7	40.2	45.4	51.6	58.3	63.4	64.7	64.1	59.7	53.5	44.6	39.3	(33)
		MCDB	41.1	43.7	50.7	61.0	68.6	74.6	76.3	75.5	67.4	57.4	47.0	41.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	12.1	15.5	20.2	20.6	20.3	20.3	20.5	19.5	15.2	9.4	8.3	(35)
		MCDBR	11.0	16.6	24.4	29.2	29.0	28.4	29.2	29.5	27.7	20.1	14.3	11.1	(36)
	5% WB	MCWBR	8.9	11.6	14.6	15.1	14.1	12.3	11.3	12.3	14.2	12.4	10.4	9.1	(37)
		MCWBR	10.7	14.7	20.9	26.9	26.0	25.2	24.8	24.9	23.9	18.1	12.9	10.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.300	0.321	0.359	0.393	0.399	0.400	0.410	0.404	0.375	0.351	0.325	0.297	(40)
		taud	2.421	2.371	2.329	2.269	2.282	2.306	2.276	2.314	2.370	2.431	2.460	2.433	(41)
	Ebn at Noon		240	261	268	270	273	273	268	264	260	246	228	224	(42)
		Edh at Noon	20	27	33	39	40	40	40	37	31	25	20	17	(43)
(44) All-Sky Solar Radiation	RadAvg		275	546	885	1345	1582	1747	1660	1452	1048	627	304	215	(44)
	RadStd		48	86	115	176	182	145	161	142	140	75	36	26	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	+0.54	+0.70	N/A	N/A	+103	+38	

Nomenclature: See separate page